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Extruded aluminum canopy with hidden fasteners

Abstract

A system for extrusion based hidden fastener canopies includes an extruded canopy support member, a facial member, a gutter cap member, a clip member, and a fastener. The extruded canopy support member has a facial surface. The facial member has at least one track, and the facial member is attached to the facial surface. The gutter cap member is connected to the track. The clip member includes a first and second edge, which are spaced apart by a length of material that corresponds to an attachment dimension of the facial member. The facial surface and an edge of the clip member receive the fastener. Both the clip member and the fastener are hidden from view.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 17/029,859, filed 23 Sep. 2020, titled “Extruded Aluminum Canopy With Hidden Fasteners,” which is a continuation-in-part of U.S. patent application Ser. No. 15/986,275, filed 22 May 2018, titled “Extruded Aluminum Canopy With Hidden Fasteners,” which issued as U.S. Pat. No. 10,907,359 on 2 Feb. 2021, which is a divisional of U.S. patent application Ser. No. 14/975,014, filed 18 Dec. 2015, titled “Extruded Aluminum Canopy With Hidden Fasteners,” which issued as U.S. Pat. No. 9,976,310 on 22 May 2018, all of which are hereby incorporated by reference for all purposes as if fully set forth herein.

BACKGROUND

1. Field of the Invention

(1) The present invention relates generally to canopies, and more specifically to a system and method for canopies that feature hidden fasteners.

2. Description of Related Art

(2) Canopies are structures that provide shade and protection from rain. Canopies are typically affixed to the sides of buildings near doors for shelter. Canopies are also used for aesthetic purposes to decorate buildings. Conventional canopies are assembled and welded together at the shop. After welding, the canopy is sanded and then painted or finished. The canopy is then shipped to the building and hoisted up to be secured to the building. Thus conventional canopies are size limited by the ability to fabricate them and move them in one-piece to the jobsite. Additionally a large canopy is difficult to hoist due to the weight and size. Furthermore, the one piece canopy cannot be resized at the jobsite as the assembly is welded together preventing on site job adjustment. Thus, there exists significant room for improvement in the art for overcoming these and other shortcomings of conventional systems and methods for canopies attached to the side of buildings.

Description

DESCRIPTION OF THE DRAWINGS

(1) The novel features believed characteristic of the embodiments of the present application are set forth in the appended claims. However, the embodiments themselves, as well as a preferred mode of use, and further objectives and advantages thereof, will best be understood by reference to the following detailed description when read in conjunction with the accompanying drawings, wherein:

- (2) FIG. 1 is a generally upward perspective view of a canopy according to the present application;
- (3) FIG. 2 is a generally downward perspective view of a canopy according to the present application;
- (4) FIG. 3 is a magnified perspective view of the canopy in FIG. 2 according to the present application;
- (5) FIG. 4 is a magnified perspective view of the canopy in FIG. 2 according to the present application;
- (6) FIG. 5A is a back view of an extruded member according to the present application;
- (7) FIG. 5B is an end view of an extruded member according to the present application;
- (8) FIG. 5C is a perspective view of an extruded member according to the present application;
- (9) FIG. 6A is a perspective view of the back of an extruded member assembly according to the present application;
- (10) FIG. 6B is a top view of an extruded member assembly according to the present application;
- (11) FIG. 6C is an end view of an extruded member assembly according to the present application;
- (12) FIG. 6D is a perspective view of the front of an extruded member assembly according to the present application;
- (13) FIGS. 7A and 7B are perspective views of the back of an extruded member assembly according to the present application;
- (14) FIG. 8 is a perspective view of the back of an extruded member assembly according to the present application;
- (15) FIG. 9 is a side section view of a canopy and a multi-piece fascia system according to the present application;
- (16) FIGS. 10-17 are side section views of alternative facial cap members according to the present application;
- (17) FIGS. 18-23 are side section views of alternative gutter cap members according to the present application;
- (18) FIGS. 24-26 are side section views of alternative clip support members according to the present application; and
- (19) FIG. 27 is a perspective view of an alternative embodiment of a clip support member according to the present application.
- (20) While the assembly and method of the present application is susceptible to various modifications and alternative forms, specific embodiments thereof have been shown by way of example in the drawings and are herein described in detail. It should be understood, however, that the description herein of specific embodiments is not intended to limit the invention to the particular embodiment disclosed, but on the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the present application as defined by the appended claims.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

- (21) Illustrative embodiments of the extruded aluminum canopy with hidden fasteners system and method are provided below. It will of course be appreciated that in the development of any actual embodiment, numerous implementation-specific decisions will be made to achieve the developer's specific goals, such as compliance with assembly-related and business-related constraints, which will vary from one implementation to another. Moreover, it will be appreciated that such a development effort might be complex and time-consuming, but would nevertheless be a routine undertaking for those of ordinary skill in the art having the benefit of this disclosure.
- (22) Referring now to FIGS. 1-4 in the drawings, a preferred embodiment of a canopy with hidden fasteners according to the present application is illustrated. Canopy **101** is comprised of a front member **105**, a back member **109**, a first end member **113**, a second end member **117**, and a deck panel **121**. Canopy **101** is installed on the sides of buildings. First the back member **109** is fastened to the surface of the building. First end member **113** and second end member **117** are coupled to the

back member by way of brackets **125**. Front member **105** is coupled to the first end member **113** and second end member **117** by way of brackets **125** thereby forming a rectangular frame. Deck panel is then sized to fit the canopy, cut to length and width and then coupled to the rectangular frame. Deck panel **121** is typically sized to leave a substantial opening over the gutter portion of the extruded member. Alternatively the deck panel is sized to only leave a small gap above the gutter portion of the extruded member to act as a leaf guard. Deck panel is comprised of interlocked panels and is watertight so that rain falling on the canopy is directed to the extruded members and drained away. Furthermore, braces **131** are installed above the deck. Turnbuckles can be attached from the surface of the building to the braces **131** to further support canopy **101**.

(23) Canopy **101** can be cut to length in a shop or in at the installation site because the extruded members are shipped unassembled from the shop to the installation site. Furthermore, because the extruded members can be cut at the site before assembly the parts for the canopy can be adjusted at the site. For example, if the measured length of the canopy needed to be reduced, the back and front extruded members can be precisely cut to reduce the length of the canopy. Conventional welded canopies require a large amount of work to adjust the dimensions of the canopy. Since the canopy is assembled at the installation site, the amount of equipment to hoist the canopy up bit by bit is less than the amount of hoisting equipment needed for conventional welded canopies.

(24) It should be apparent that the canopy **101** does not have fasteners viewable from underneath the canopy. The hidden fasteners of canopy **101** increase the aesthetic appeal of the canopy. Furthermore, the hidden fasteners require less finishing work and over time any corrosion due to a reaction between the fasteners and the extruded members is hidden from view.

(25) Referring now also to FIG. 5A-5B in the drawings, a preferred embodiment of an extruded member according to the present application is illustrated. Extruded member **201** is comprised of aluminum and extruded into a shape configured for use with hidden fasteners. Extruded member **201** has a gutter portion **205**, a tracked portion **207**, and a member length. Tracked portion **207** is comprised of at least one track **209** sized to retain a head of a bolt and prevent the bolt from twisting. The head of the bolt can be slid up and down the track however a width of the track is slightly larger than the width of the head of the bolt to prevent the bolt from rotating relative to the track. As illustrated the tracked portion further comprises of a second track **211** sized to retain a head of a bolt and prevent the bolt from twisting. It should be apparent that more tracks are contemplated by this application. Furthermore, while both tracks are illustrated outside the gutter portion **205**, it should also be apparent that tracks inside the gutter portion are contemplated by this application. A benefit of locating the tracks in the tracked portion **207** is access to the tracks once the deck panel is attached. Both the first track **209** and the second track **211** run the entire length of the extruded member **201**. It should be apparent that escape slots may be located in the tracks to facilitate removal and insertion of fasteners from the extruded member without being at the end of the extruded member. Escape slots are sized that the head of the fastener can be inserted or removed from the middle of the extruded member. Furthermore, there can be a plurality of escape slots located a set distance apart, for example every 1 foot.

(26) Referring now also to FIG. 6A-6D in the drawings, a preferred embodiment of an extruded member corner assembly according to the present application is illustrated. Assembly **301** is comprised of a first extruded member **305**, a second extruded member **307**, a bracket **311**, and a plurality of fasteners **313**. The plurality of fasteners is preferably comprised of bolts and nuts. First extruded member **305** is trimmed at a forty-five degree angle at the assembled end. Second extruded member **307** is trimmed at a forty-five degree angle at the assembled end.

(27) To assemble the right angle assembly **301**, the head of the bolts are slid into each track of the extruded members. The bracket **311** is then coupled to the bolts and secured by nuts. A colored matched filler, such as epoxy or silicone, is applied between the extruded members to fill any small gaps. Bracket **311** as illustrated is a right angle bracket and thereby forms a right angle assembly. Other shapes and angles are possible based upon the shape of the bracket. The finished corner as

shown in FIG. 6D results in a canopy without exposed fasteners viewable from underneath the assembled canopy.

(28) Referring now also to FIG. 7A-7B in the drawings, a preferred embodiment of an extruded member straight assembly according to the present application is illustrated. Straight assembly **401** is comprised of a first extruded member **405**, a second extruded member **407**, a bracket **411**, and a plurality of fasteners **413**. The plurality of fasteners is preferably comprised of bolts and nuts. First extruded member **405** is trimmed at a ninety degree angle at the assembled end. Second extruded member **407** is trimmed at a ninety degree angle at the assembled end.

(29) To assemble the assembly **401**, the head of the bolts are slid into each track of the extruded members. The bracket **411** is then coupled to the bolts and secured by nuts. A colored matched filler, such as epoxy or silicone, is applied between the extruded members to fill any small gaps. Bracket **411** as illustrated is a straight bracket and thereby forms a straight assembly. Other shapes and angles are possible based upon the shape of the bracket. Assembly **401** allows users to create canopies of various lengths without limitations based upon the length of the extruded members.

(30) It should be apparent that a canopy can be comprised of multiple right angle assemblies **301**. Furthermore, it should be apparent that multiple right angle assemblies **301** can be joined together by straight assemblies **401** to form a canopy. Additionally, the parts of the canopy can be preassembled at a shop and shipped to the installation site. For example, two right angle assemblies **301** can be formed at a shop, shipped to the installation site, and then combined to form a canopy. Alternatively, four right angle assemblies **401** can be preassembled at the shop, shipped to the installation site, and then combined by assemblies **401** to form a canopy.

(31) The system and method described above solves problems inherent in conventional canopies. First, conventional canopies are welded together and therefore require cutting, welding, sanding, and finishing to repair a damaged canopy. The improved system comprised of extruded members can be disassembled, damaged part replaced, and reassembled without the need to weld, sand, or refinish the canopy. Second, conventional canopies have exposed fasteners viewable from the front and underside of the canopy. The improved system hides the fasteners inside the canopy where they are less visible. Third, conventional systems are assembled and assembled at a shop and then trucked to the installation site, thereby limiting the size of canopy based at least upon the ability to move the conventional welded canopy. The improved system allows a canopy to be shipped unassembled and then assembled at the installation site. Furthermore, it is safer to hoist the canopy a piece at a time instead of a heavier completely assembled canopy.

(32) A method for assembling a canopy having hidden fasteners; providing a first member having at least two tracks; providing a second member having at least two tracks; providing a bracket; providing a first set of fasteners; providing a second set of fasteners; sliding the first set of fasteners into the at least two tracks of the first member; sliding the second set of fasteners into the at least two tracks of the second member; and coupling the first member to the second member with the bracket and both the first set of fasteners and the second set of fasteners. Furthermore, by sizing the at least two tracks of the first member to prevent the first set of fasteners rotating relative to the first member; and by sizing the at least two tracks of the second member to prevent the second set of fasteners rotating relative to the first member the sets of fasteners can be retained by the tracks without the using shims or additional members inside the tracks other than the bolt or fastener. Because the members are not welded together they can be disassembled and reassembled to replace defective members by providing a third member having at least two tracks; uncoupling the first member to the second member with the bracket and both the first set of fasteners and the second set of fasteners; sliding the first set of fasteners out of the at least two tracks of the first member; sliding the first set of fasteners into the at least two tracks of the third member; and coupling the third member to the second member with the bracket and both the first set of fasteners and the second set of fasteners.

(33) Referring now also to FIG. 8 in the drawings, an alternative embodiment of an extruded

member portion assembly according to the present application is illustrated. Assembly **501** is comprised of a first extruded member **505**, a second extruded member **507**, a third extruded member **509**, a bracket **511**, and a plurality of fasteners **513**. The plurality of fasteners is preferably comprised of bolts and nuts. First extruded member **505** is trimmed at a forty-five degree angle at a welded end. Second extruded member **507** is trimmed at a forty-five degree angle at a welded end. First extruded member **505** is welded to second extruded member **507** to form a 90 degree angled corner. Third extruded member **509** is coupled to the welded corner by bracket **511**. This embodiment allows a user to fabricate four corners in the shop and take them out to the job site. Once there the straight extruded members can be attached with the hidden fasteners to form a canopy.

(34) Referring now also to FIG. **9** in the drawings, a multi-piece fascia system **901** is illustrated. Multi-piece fascia system **901** is comprised of an extruded canopy support member **601** having a gutter portion **605**, a track portion **607**, and one or more facial surfaces **650a**, **650b**, **650c**, **650d** with a facial member **951**, a gutter cap member **959**, and a clip support member **961** attached to one or more facial surfaces **650**. Multi-piece fascia system **901** is attached to extruded canopy member **601** using one or more fasteners **913a**, **913b**, **913c**, **913d**, **913e**, **913f**, and **913g**. Although multi-piece fascia system **901** is depicted as attached to an extruded canopy member, other uses are contemplated by the present application. For example, multi-piece facial system **901** may be attached or integrated with a deck pan, a fence rail, a truss, an awning, or a portion thereof.

(35) In a preferred embodiment, gutter cap **959** includes an opening or a track that fits around gutter portion **605** of extruded canopy member **601**. The dimensions are such that gutter cap **959** leaves a slight airgap at least on one side (bottom) between an edge of gutter cap **959** and an edge of extruded canopy member **601**. The airgap includes an adhesive strip (below).

(36) In a preferred embodiment, clip support member **961** is a plurality of sheared extruded bracket portions evenly spaced across extruded canopy member **601**. A number of brackets included in the plurality of bracket portions is dependent at least on a length of extruded canopy member **601** and a desired structural stability. Alternatively, clip support member **961** is a single, continuous extruded bracket commensurate in length with the length of extruded canopy member **601**. It is noted that facial member **951** and gutter cap member **959** may also comprise a plurality of sheared portions, but are preferably continuous lengths in order to provide unitary structural and aesthetic features.

(37) In a preferred embodiment, clip support member **961** has a first edge **963a** and a second edge **963b**. First and second edges **963a**, **963b** are spaced apart by a length of material **965**, where length of material **965** corresponds to an attachment dimension **967** of facial member **951**.

(38) In a preferred embodiment, multi-piece fascia system **901** is attached to extruded canopy member **601** using a combination of adhesive and hidden fasteners. For example, one or more adhesive portions **913c**, **913d**, **913e**, and **913f** adhere multi-piece fascia system **901** to an exterior surface of extruded canopy member **601**. Preferably, adhesive portions **913c**, **913d**, **913e**, and **913f** are adhesive tape portions. In other embodiments, adhesive **913c**, **913d**, **913e**, and **913f** is a continuous adhesive strip or adhesive spray adhered between extruded canopy member **601** and multi-piece fascia system **901**. A screw **913a** secures clip support member **961** to a top facial surface **650b** of extruded canopy member **601**. A rivet **913b** secures a flange of clip support member **961** to an attachment tab of facial member **951**. Both screw **913a** and rivet **913b** are visually imperceptible relative to the front facial surface from a front perspective and from a generally upward perspective view of a canopy incorporating the multi-piece fascia system **901**. Screw **913a** is imperceptible relative to upward and downward perspectives.

(39) In a preferred embodiment, multi-piece fascia system **901** includes at least three bent or extruded aluminum members. Bent aluminum members are made from a flat pattern from a single unitary piece of material and may include fold or bend lines to indicate how to form the member. In other embodiments, the number of members and the type of material varies depending on expected use, weather conditions, desired appearance, and combinations thereof. For example, an extruded

plastic or polymer may be used, reducing the weight of the facial system and allowing clip support member **961** to be extended, removed, or exchanged. In a preferred embodiment, adhesive is not applied between clip support member **961** and facial member **951**, saving on adhesive material. In other embodiments, adhesive is applied between clip support member **961** and facial member **951**. (40) In a preferred embodiment, the ratio of facial member **951** height to canopy extruded member **601** height is $3/2$. For example, facial member **951** is one of a four-inch, six-inch, eight-inch, or 12-inch high extruded member, while canopy extruded member **601** is one of a two-and-two-thirds-inch, four-inch, six-inch, or eight-inch high extruded member. In other embodiments, the ratio increases above $3/2$, and multi-piece facial system **901** incorporates a structurally reinforced clip support member **961** and/or structurally reinforced facial member **951**. In yet other embodiments, the ratio decreases below $3/2$, clip support member **961** becomes an optional component of the multi-piece facial system **901**, and a shear and/or break feature of at least the facial member **951** is incorporated.

(41) In a preferred embodiment, clip support member **961** is C-shaped having a thickness that is the same as the thickness of facial member **951**. In other embodiments, clip support member **961** varies in thickness across its surface and/or relative to the thicknesses of the other multi-piece facial system members. In yet other embodiments, clip support member **961** includes additional folds, flutes, channels, bends, surface hardening, or combinations thereof, in order to provide additional support to facial members having large dimensions. In other embodiments, facial member **951** and the gutter cap member **959** may also have additional folds, flutes, channels, bends, varying thicknesses, surface hardening, or combinations thereof, in order to provide additional support. In a preferred embodiment, the flanges of the C-shaped clip support member **961** form a right angled track for retaining the screw **913a** in such a way that the screw may be rotated and removed. In other embodiments, at least a top flange of the clip support member **961** is not at a right angle, but rather at an acute angle with the corresponding flange of facial member **951** also being at a corresponding acute angle.

(42) In a preferred embodiment, facial member **951** includes a C- or inverted F-shaped track **957**. The C- or inverted F-shaped track **957** overlaps and integrates with the gutter cap member **959**. The overlap is sufficiently sized to accommodate any protruding fasteners the multi-piece facial system **901** is intended to cover up. For example, if a $1/4$ inch head of a bolt protrudes from the face of a canopy, at least the outermost flange (bottom flange in FIG. 9) of the track **957** is more than $1/4$ inches in length, such that the underlying canopy structure is still facially covered by the overlap of facial member **951** integrated with the gutter cap member **959**. Additional factors that determine the length of the flange of the track **957** and the corresponding air gap size include, but are not limited to, whether facial member **951** is flexible or rigid, a location of the fasteners to be covered up on the facial surface of the canopy, an amount of material used to form facial member **951**, and a desired aesthetic appearance created by the overlapping feature of track **957**. In at least one embodiment, track **957** includes a ridge **913g** and gutter cap **959** includes a corresponding ridge to provide slight lateral support and prevent an unintentional separation of facial member **951** and the gutter cap **959**.

(43) In a preferred embodiment, the gutter cap **959** includes at least two right angles, sized to encase, fit over, snap onto, clip to, or removably attach with the gutter portion **605** of extruded canopy member **601**. The adhesive **913e** and/or ridge **913g** prevents unwanted removal of gutter cap **959** from gutter portion **605**.

(44) In an alternative embodiment, multi-piece fascia system **901** is a single, unitary cap member (not shown), which includes an attachment tab to connect to an edge of the clip member and a flange to snap onto a facial surface (e.g., surface **650c**) of a gutter portion of a canopy support member. The single, unitary cap member connects the facial member, the clip support member, and the gutter cap member as a single, unitary structure (not shown) that is either bent or extruded to take its form. The fastener, together with the single unitary cap structure, comprises two

components of an alternative multi-piece facial system.

(45) Referring now to FIGS. **10-17** in the drawings, alternative embodiments of section views of facial members are depicted. For example, referring now to FIG. **10**, a facial member **1051** includes a first track member **1057a**, second track member **1057b**, and a third track member **1057c**.

Preferably, at least one of the track members **1057** is formed as an attachment tab, meaning it receives a fastener to connect it with a flange or tab of gutter cap **959** or clip support member **961**. In some embodiments, two or all of the track members **1057** are formed as attachment tabs. Track members **1057a** and **1057c** are substantially equal in the length of the side comprising the track member, while track member **1057b** has a greater length.

(46) Referring now to FIG. **11**, facial member **1151** includes first track member **1157a** and second track member **1157b**. First track member **1157a** includes a ridge element **1113**. Alternatively, track member **1157b** or both track members include ridge element **1113**.

(47) Referring now to FIG. **12**, facial member **1251** includes track members **1257a** and **1257b** that are substantially different in length. For example, track member **1257a** is less than or equal to half the length of track member **1257b**. Track members **1257a** and **1257b** are separated by a length of material that is greater than the lengths of one or both of track members **1257a** and **1257b**.

(48) Referring now to FIG. **13** in the drawings, track members **1357a** and **1357b** are separated by a length of material that is less than the length of one or both of the track members, and are also connected to a length of material that is greater than or equal to a facial dimension of an extruded canopy support member, such as canopy support member **601**.

(49) Referring now to FIG. **14** in the drawings, facial member **1451** includes a first curved track member **1457a** and a second linear track member **1457b**. Alternatively, **1457b** may be curved and **1457a** linear, or both are curved, relying on the curvature of at least one of the track members to retain the facial member onto a facial surface of the extruded canopy support member.

(50) Referring now to FIG. **15** in the drawings, facial member **1551** includes first track member **1557a**, and second and third oppositely oriented track members **1557b** and **1557c**. First track member **1557a** is connected to second and third track members **1557b**, **1557c** by a length of material that is less than or equal to a facial dimension of a facial surface of an extruded canopy support member, such as canopy support member **601**. This type of facial member is used together with a gutter cap member that overlaps portions of the same facial surface as facial member **1551** to make up for any surface not covered by facial member **1551**. Facial member **1551** may also optionally include a ridge **1513**.

(51) Referring now to FIG. **16** in the drawings, facial member **1651** is shaped similar to facial member **1351**, except that facial member **1651** includes one or more additional structural reinforcing features and/or fastener elements. For example, first track member **1657a** includes ridge **1613** that fits with another ridge of a gutter cap flange. In some embodiments, the structural reinforcing and/or fastening feature includes a fold, a bend, a groove, an extra layer of material, or other similar features. Alternatively, second track member **1657b** includes the structural reinforcing or fastening feature.

(52) Referring now to FIG. **17** in the drawings, facial member **1751** is shaped similar to facial member **1551**, except that there is no latitudinal track member connected to first track member **1757a** and second track member **1757b**. Additionally, the track members **1757a**, **1757b** are connected to a length of material that is greater than or equal to a facial dimension of the facial surface of the extruded canopy support member, such as canopy support member **601**.

(53) Referring now to FIGS. **18-22** in the drawings, section views of alternative embodiments of gutter cap members are depicted. For example, referring now to FIG. **18**, gutter cap member **1859** includes first flange **1857a** and second flange **1857b** connected by a length of material that is commensurate in length (e.g., greater than or equal to) with a facial dimension of a facial surface of a gutter portion of an extruded canopy support member, such as facial surface **650c** of gutter portion **605**. It is important to note that a side length of first flange **1857a** may be shortened, such that all

that remains of the flange is a gripping lip surface that is about 0.1-0.3 inches in length. Preferably, an end of second flange **1857b** connects to or is seated within one or more track members of a facial member, such as facial member **951**. Preferably, at least one flange of the gutter cap member is configured as an attachment tab.

(54) Referring now to FIG. **19** in the drawings, gutter cap member **1959** includes a single flange **1957** connected to a length of material that is greater than or equal to a facial dimension of a facial surface of a gutter portion of an extruded canopy support member, such as facial surface **650c** of gutter portion **605**. Although not depicted, flange, the length of material, or both may include structural reinforcement features and/or fastening features.

(55) Referring now to FIG. **20** in the drawings, gutter cap member **2059** includes a single flange **2057**. Flange **2057** is connected to a length of material that is less than or equal to a facial dimension of a facial surface of a gutter portion of an extruded canopy support member, such as facial surface **650c** of gutter portion **605**. Flange **2057** includes ridge **2013**, or another structural reinforcing or attachment feature.

(56) Referring now to FIG. **21** in the drawings, gutter cap member **2159** includes first flange **2157a** and second flange **2157b** connected to a length of material that is greater than or equal to a facial dimension of a facial surface of a gutter portion of an extruded canopy support member, such as facial surface **650d** of gutter portion **605**.

(57) Referring now to FIG. **22** in the drawings, gutter cap member **2259** having a single flange **2257** is depicted. Single flange is connected to a length of material that is greater than a facial dimension of a facial surface of a gutter portion of an extruded canopy support member, such as facial surface **650c** of gutter portion **605**. Flange **2257** and the length of material include ridges **2213a**, **2213b**, or similar structural reinforcing and/or attachment features.

(58) Referring now to FIG. **23** in the drawings, gutter cap member **2359** includes first flange **2357a** and second flange **2357b** connected to a length of material that is greater than or equal to a facial dimension of a facial surface of a gutter portion of an extruded canopy support member, such as facial surface **650d** of gutter portion **605**. Preferably, the length of material is greater than the facial dimension to provide air gaps and adhesive between the gutter cap member and the gutter cap portion **605**. Preferably, flange **2357b** covers a portion of facial surface **650a** and connects to a track member of a facial member, such as facial member **1551**. Flange **2357b** includes ridge **2313**, or another structural reinforcing or attachment feature, which preferably interlocks with ridge **1557b**.

(59) Referring now to FIGS. **24-26** in the drawings, section views of alternative embodiments of clip support members are depicted. For example, referring now to FIG. **24**, clip support member **2461** includes first edge **2457a** and second edge **2457b** connected by a length of material, such as length **965** (FIG. **9**), that is above a facial surface of a track portion of an extruded canopy support member, such as facial surface **650a** of track portion **607**. It is important to note that a side length of first edge **2457a** may be shortened, such that all that remains of the attachment tab is a lip surface that is about 0.01-0.1 inches in length. Preferably, first edge **2457a** is greater than or equal to a width dimension (e.g., diameter) of a rivet, such that a rivet may be received in the attachment tab. Second edge **2457b** connects to or is seated adjacent or on top of a facial surface of the extruded canopy support member, such as facial surface **650b**. Preferably, the clip members depicted in FIGS. **24-26** are formed by extrusion. Alternatively, they are formed by folding or bending sheet metal. Preferably, an attachment dimension **967** of a facial member depends on length of material **965** (FIG. **9**), meaning that facial member may retrofit or attach to any canopy support or a deck panel of any dimension due to the structural support provided by the clip member and its length of material.

(60) Referring now to FIG. **25**, clip support member **2561** includes edge **2557a** and second edge **2557b** connected by a length of material that is above a facial surface of a track portion of an extruded canopy support member, such as facial surface **650a** of track portion **607**. It is important

to note that a side length of second edge **2557b** may be lengthened, such that greater than 50% of facial surface **650b** is covered by edge **2557b**. First edge **2557a** includes a ridge **2513**, or structural attachment feature to interlock or interface with a track member of a facial member, such as ridge **1113** of facial member **1151**.

(61) Referring now to FIG. **26** in the drawings, clip support member **2661** includes a first curved edge **2657a** and a second linear edge **2657b**. Alternatively, **2657b** may be curved and **2657a** linear, or both are curved, relying on the curvature of at least one of the edge to retain the track members of a facial member. For example, curved edge **2657a** may interface with curved track member **1451a** of facial member **1451**. Preferably, a length of second edge **2657b** is greater than or equal to a diameter of a screw head, such that a screw or a similar fastener is received into second edge **2657b**.

(62) Referring now to FIG. **27** in the drawings, a perspective view of an alternative embodiment of a clip support member **2761** is depicted. Clip support member **2761** includes first edge **2757a** and second edge **2757b**, first ridge member **2713a**, and second ridge member **2713b**. Fold lines separate and differentiate first and second ridge members **2713a** and **2713b**. Preferably, clip support member **2761** is formed from folding or bending aluminum sheet metal, and is one of multiple clip support members that are positioned lengthwise along a canopy support member. Clip support member **2761** optionally includes multiple openings or slots **2711** for receiving a fastener, such as screw **913a** or rivet **913b**.

(63) A method of assembling a canopy having a multi-piece fascia system includes assembling the canopy using extruded canopy members, as discussed above. The multi-piece fascia system is formed according to the dimensions of the assembled canopy. The formation of the multi-piece fascia system includes coloring, applying indicia or graphics, bending (if necessary), and/or obtaining specified unconventional member lengths. The coloring of the multi-piece fascia system can include, but is not limited to, painting, dying, extrusion coating, and/or mill or mechanical finishing. Adhesive strip is applied to one or more exterior facial surfaces of the assembled canopy, or to an interior surface of the multi-piece facial system. The facial system is attached, having its interior surface abutting the adhesive strip(s) and portions of the exterior facial surface of the canopy. In a preferred embodiment, the adhesive is a tape. In other embodiments, the adhesive is an epoxy. A solvent, such as acetone, may be used to clean surfaces before applying adhesive.

(64) The multi-piece facial system fasteners are secured, providing additional support from the clip support member and thereby enabling relatively large facial dimensions. In a preferred embodiment, the attachment of the clip support member and securing of fasteners is before adhesive is applied. In other embodiments, the attachment of the clip support member and securing of the facial system fasteners occurs after adhesive is applied. Due to the lightweight, stable materials used, assembly can occur at ground level, and the canopy hoisted into position with the multi-piece facial system attached. However, in a preferred embodiment, the multi-piece facial system is attached piece-wise after the canopy is assembled piece-wise onto an exterior wall of a building.

(65) The system and method described above solves problems inherent in conventional canopies, canopy coatings, and canopy finishes. First, conventional canopies use corrosive material, are painted, or otherwise finished to avoid corrosion. The improved system comprised of pre-finished multi-piece facial members can be disassembled, damaged part replaced, and reassembled without the need to coat, finish, weld, sand, or refinish the canopy. Second, conventional canopies have exposed fasteners viewable from the front and underside of the canopy. The improved system hides the fasteners inside the canopy where they are less visible. Third, pre-existing canopies may be retrofitted with the multi-piece facial system to obtain the similar advantage of having seams and fasteners hidden by attachment of the multi-piece facial system without removing the pre-existing canopy from the wall. Fourth, conventional systems do not include gutter channeling, and if they do, the gutters require coating and/or finishing. The improved system provides gutter channeling

features that do not require coating and/or finishing. Fifth, conventional canopies require repainting, refinishing, or removing the canopy to change the color and appearance. The improved multi-piece facial system enables changes to appearance and color without repainting, refinishing, or removing canopies. Sixth, auxiliary and cosmetic components of conventional canopies are welded into place, requiring finishing after these components are attached to the canopy. The improved facial system is secured to the canopy without requiring welding. Because welding is not required, additional finishing options, such as the use of polymers, plastics, and electronics are made available. Seventh, conventional systems are limited to customary lengths, such as 10-foot sections. In the improved system, continuous lengths of 15-, 20-, or 30-foot sections, or more, are made available.

(66) It is apparent that a system with significant advantages has been described and illustrated. The particular embodiments disclosed above are illustrative only, as the embodiments may be modified and practiced in different but equivalent manners apparent to those skilled in the art having the benefit of the teachings herein. It is therefore evident that the particular embodiments disclosed above may be altered or modified, and all such variations are considered within the scope and spirit of the application. Accordingly, the protection sought herein is as set forth in the description. Although the present embodiments are shown above, they are not limited to just these embodiments, but are amenable to various changes and modifications without departing from the spirit thereof.

Claims

1. A facial system for extrusion based hidden fastener canopies, comprising: an extruded canopy support member, having: at least one outward facing facial surface; and a gutter member; a facial member having at least one track, the facial member attached to the at least one outward facing facial surface of the extruded canopy support member; an external gutter cap member connected to the at least one track; a clip member, having: a first edge; and a second edge; and a fastener; wherein the first edge and the second edge of the clip member are spaced apart by a length of material, the length of material corresponding to an attachment dimension of the facial member; wherein the clip member is connected to the at least one outward facing facial surface using the fastener; wherein the clip member and the fastener are hidden from view; wherein the external gutter cap member encloses the gutter member; and wherein a longitudinal length of the facial member runs along a plane parallel with the extruded canopy support member, such that the longitudinal length of the facial member extends along and wraps around a longitudinal length of the extruded canopy support member.
 2. The system according to claim 1, wherein one of the gutter member and the facial member further comprises: an attachment edge.
 3. The system according to claim 1, wherein the external gutter cap member is separable from the facial member.
 4. The system according to claim 1, wherein the at least one outward facing facial surface of the extruded canopy support member is one of a plurality of outward facing facial surfaces of the extruded canopy support member.
 5. The system according to claim 4, wherein the fastener is one of a plurality of fasteners.
 6. The system according to claim 5, wherein each outward facing facial surface of the plurality of outward facing facial surfaces receives a fastener of the plurality of fasteners.
 7. The system according to claim 5, wherein a fastener of the plurality of fasteners comprises an adhesive strip.
 8. The system according to claim 7, wherein more than one of the fasteners of the plurality of fasteners comprises discrete adhesive strip segments.
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